

Grade 11 Maths Unit 5

Unit Outcomes

By the end of this unit, students will be able to:

- Know basic concepts and procedures about vectors and operation on vectors.
- Know specific facts about vectors.
- Apply principles and theorem about vectors in solving problems involving vectors.

5.1 Revision on Scalars and Vectors

Competencies

- define a scalar quantity;
- identify the everyday application of scalars;
- define a vector quantity;
- identify the everyday application of vector; and
- describe the difference between a vector and a scalar quantity.

5.2 Representation of Vectors

5.2.1 Components of vectors

Competencies

- represent vector by different notions; and
- Resolve a given vector in to two components.

5.2.2 Addition and Subtraction of Vectors

Competencies

- determine the sum of two or more vectors;
- determine the difference of two vectors;

5.2.3 Multiplication of Vectors by Scalars

Competencies

- multiply a vector by a scalar.

5.2.4 Unit Vectors and Norms of Vectors

Competencies

- use unit vectors to determine the column representation of a given vectors.

5.2.5 Norms of Vectors

Competencies

- determine the magnitude of a vector.

5.3 Vectors Product

5.3.1 Scalar (inner or dot) product

Competencies

- find the scalar product (inner product, of two vectors; and
- describe some properties of scalar product of vectors.

5.3.2 Vectors in Three-dimensional Space

Competencies

- Know the concepts of coordinate of a point in space;
- Locate the points in space with the reference of axes x , y and z ; and
- Associate vectors in three-dimensional space with cross product

5.3.3 Cross Product

Competencies

- Associate determinant of a matrix with cross product; and

- Use cross product of a vector to find area of parallelogram

5.4 Application of Scalar and Cross Product

Competencies

- use cross product of a vector to find area of parallelogram

5.5 Application of Vector

5.5.1 Vectors and Lines

Competencies

- apply vectors to solve problems on geometry, algebra, mechanics and other related problems.

5.5.2 Vector and Circles

Competencies

- write equation of a circle by applying vectors; and
- determine the equation of the tangent line to a circle using vectors.

5.5.3 Equations of Tangent Lines to a Circle

Competencies

- determine the equation of the tangent line to a circle using vectors.

5.6 Applications

Competencies

- Use the concept of vectors to solve different real-life problems; and
- Use the concept of vectors to avoid car accidents